```
$ node server.js
✓ data/portfolio.json loaded · cache warmed · alert timer armed
✓ running at http://localhost:8000 (bound to 127.0.0.1)
```

FinSight Terminal

The **system spec**: architecture, data, every API route, every upstream call and every rule, as built and taken straight from the source.



LOC 10,783 DEPS 0 ■ ROUTES 58 ▲ TESTS 96/96 ▲ SOURCES 5 API KEYS 0 ■ P

DOC	VERSION	SOURCE	ISSUED
System spec	1.0.0	037db01 · 18 Oct 2026	24 Oct 2026

§01 · AT A GLANCE

One process. One folder. Zero keys.

01

A Bloomberg-style portfolio terminal for someone who **funds in SGD and invests in USD**. It splits every gain into **stock move** and **currency move**.

10.8k	0	58	5	96/96
LINES OF CODE	DEPENDENCIES	API ROUTES	DATA SOURCES	TESTS PASS

HOUSE RULES

01 — Local-first
Loopback only. Data never leaves `data/`.

02 — Zero deps
Node built-ins and plain browser JS. No build step.

03 — No keys
Free, keyless providers only.

04 — Degrade gracefully
Last good value beats an error.

05 — Records only
Never places an order.

WHAT IT DOES

- **P&L** — **Live portfolio** — Quotes, P&L, weights
- **FX** — **Stock vs FX split** — Per-lot buy-time rate
- **VaR** — **FX risk cockpit** — Exposure, drift, 1-wk VaR
- **TWR** — **Real returns** — Trade replay, XIRR, benchmark
- **DIV** — **Dividends** — Net of withholding
- **LED** — **Ledger + undo** — 25 steps, realized by year
- **CHT** — **Pro charts** — 24 indicators, drawings
- **STR** — **Strategy library** — 5 fixed-rule backtests
- **JRN** — **Trade journal** — Plans, exits, R stats
- **CAL** — **Macro + calendar** — FRED, World Bank, FOMC
- **ALR** — **Alerts** — Price, line, indicator
- **IMP** — **Broker import** — IBKR, moomoo, Tiger, CSV

BUILT ON

Browser	HTML · CSS · JS · Canvas
Node.js ≥ 18	http · fetch · crypto
JSON files	atomic rename
macOS	launchd · osascript

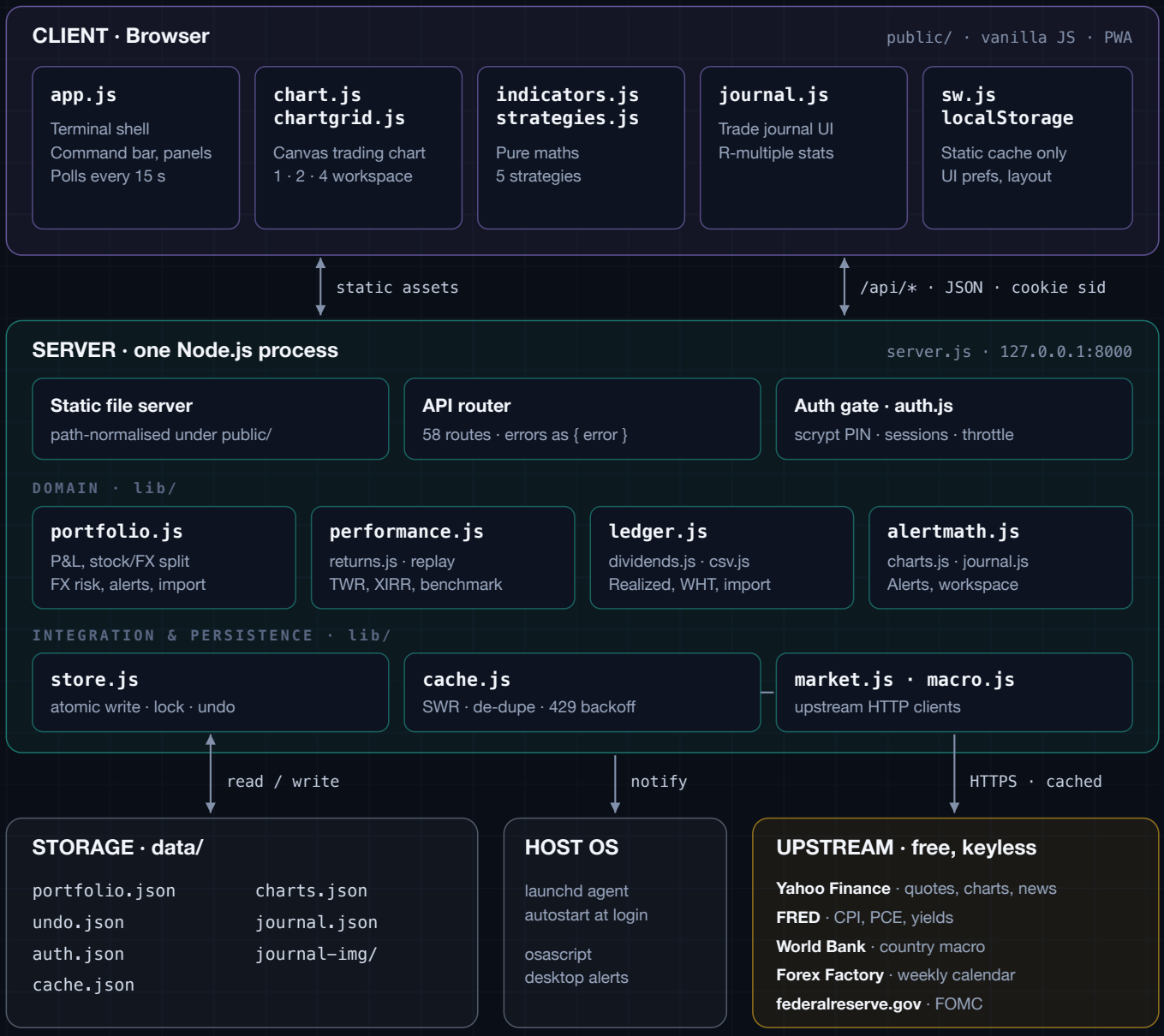
INSIDE

- Architecture 03
- Life of a request 04
- Code map 05
- Data model 06
- API route map 07
- Upstream & cache 08
- The maths 09
- Security 10
- Run it 11
- Known gaps 12
- Commands & hooks 13

§02 · ARCHITECTURE

Three tiers, one machine.

The browser only talks to `localhost`. The server is the only thing that reaches the internet, and every call goes through one cache.



No CORS, one cache
The browser never calls a provider. All traffic is pooled and rate-limit-safe.

Nothing on the LAN
Bound to 127.0.0.1. Change `HOST` deliberately or not at all.

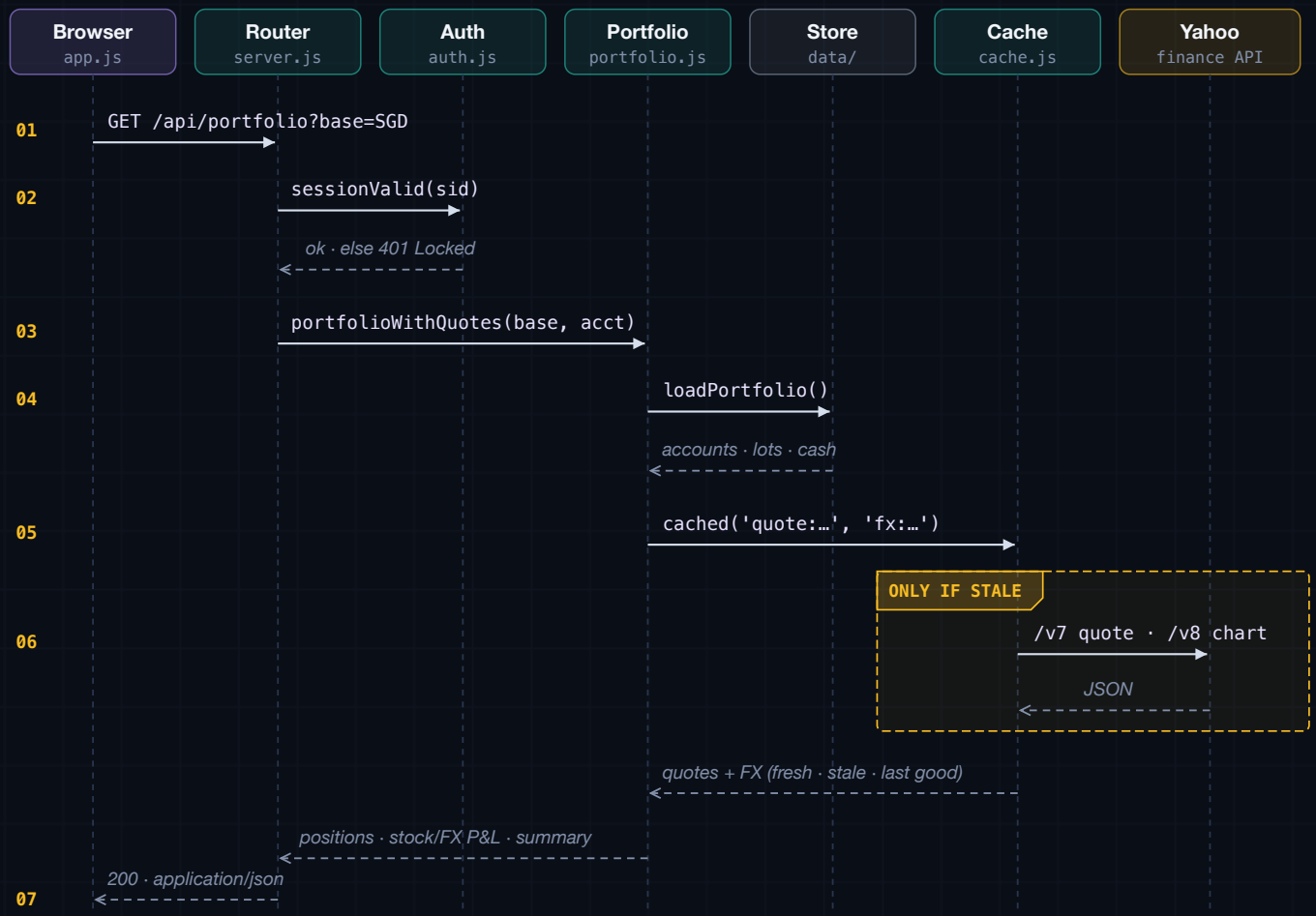
Alerts without a tab
The timer runs in the server. launchd keeps the server up.

§03 · LIFE OF A REQUEST

Every 15 seconds, this happens.

03

The portfolio refresh is the busiest path. It checks the session, reads the file, pulls quotes through the cache and returns the stock/FX split.



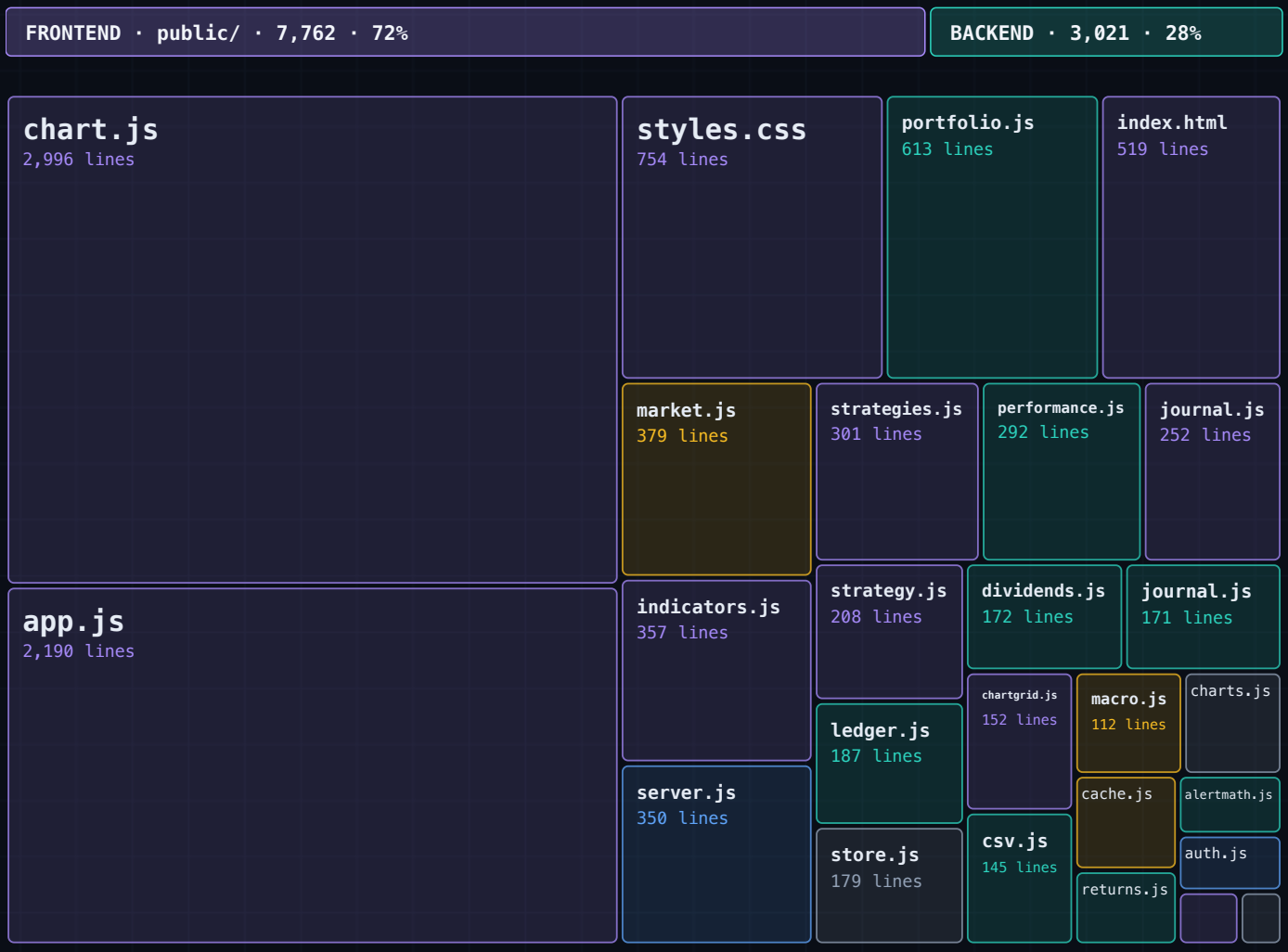
THE HEARTBEAT: WHAT POLLS, AND HOW OFTEN (LOG SCALE)



§04 · CODE MAP

Where the 10,783 lines live.

Each block's area matches its line count. Almost three-quarters is frontend, and most of that is the chart. The server is a thin router over pure `lib/` modules.



DOMAIN

portfolio trades, P&L, FX risk, alerts · **performance** replay · **returns** TWR, XIRR · **ledger** realized · **dividends** WHT, forward · **csv** broker parsing · **alertmath** line and indicator values · **journal** R stats

INTEGRATION STORE

market Yahoo client · **macro** FRED, World Bank, calendar · **cache** SWR + backoff · **store** load, migrate, lock, undo · **charts** drawings, layouts · **config** paths, port · **auth** PIN, sessions

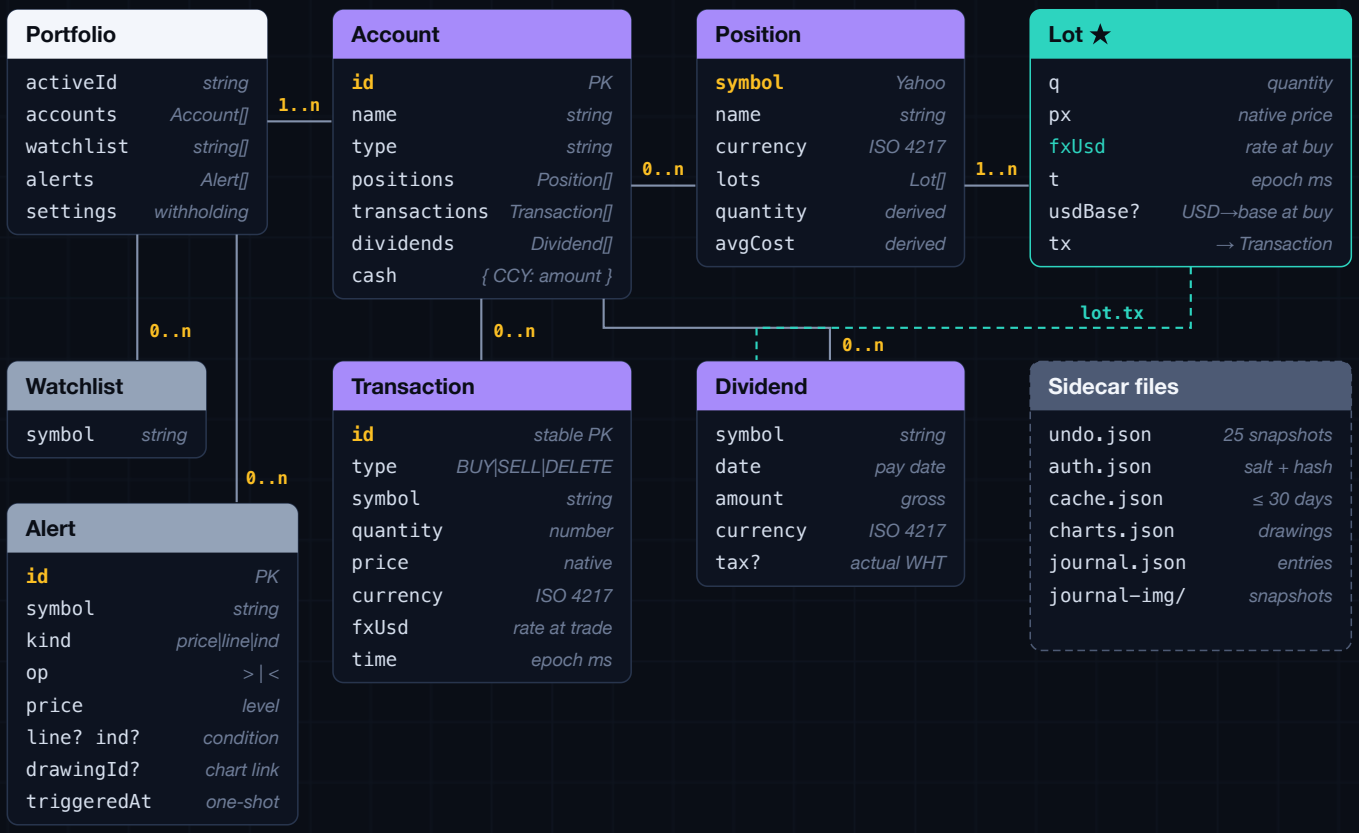
FRONTEND

chart canvas chart, drawings, compare · **app** shell, commands, wizard · **indicators** pure maths · **strategies / strategy** backtests · **chartgrid** multi-chart · **journal** UI · **sw** static cache

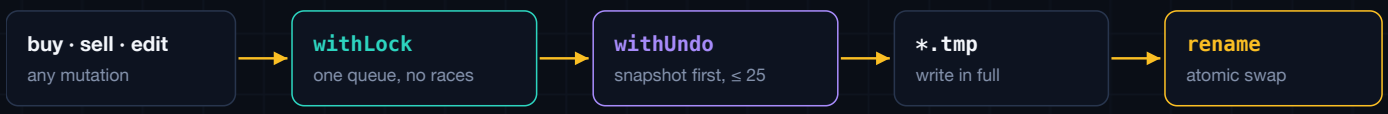
§05 · DATA MODEL

The database is a folder.

No DB server. `portfolio.json` holds everything that matters. Every position is a stack of **lots**, and each lot remembers the FX rate on the day it was bought.



EVERY WRITE TAKES THE SAME PATH



Corrupt file on load → renamed to `*.corrupt.<ts>.json`, never overwritten. A locked function never calls another locked one (that would deadlock).

§06 · API ROUTE MAP

58 routes, one page.

06

JSON in, JSON out. Add `?base=SGD` and `?account=id|ALL` to portfolio routes. Once a PIN is set, everything except `auth` and `health` needs the `sid` cookie.

GET · read | POST · change | GET + POST | 401 when locked | 404 { error: "Unknown endpoint" }

SYSTEM · AUTH 6

GET	/api/health	public
GET	/api/auth/status	
POST	/api/auth/setup	4–12 digits
POST	/api/auth/login	throttled
POST	/api/auth/logout	
POST	/api/auth/change	

MARKET 6

GET	/api/quote	?symbols=
GET	/api/history	OHLCV
GET	/api/search	?q=
GET	/api/markets	overview
GET	/api/news	?symbol=
GET	/api/categories	sectors

MACRO 3

GET	/api/fred	?series=
GET	/api/worldbank	?indicator=
GET	/api/calendar	FOMC + week

ACCOUNTS 4

GET	/api/accounts	
POST	/api/accounts/add	
POST	/api/accounts/rename	
POST	/api/accounts/remove	

PORTFOLIO 12

GET	/api/portfolio	live P&L
POST	/api/portfolio/buy	date → FX
POST	/api/portfolio/sell	realized
POST	/api/portfolio/delete	
POST	/api/portfolio/import	CSV, IBKR
POST	/api/portfolio/reset	
GET	/api/portfolio/performance	TWR
POST	/api/cash	
GET	/api/fx-risk	VaR
GET	/api/dividends	net WHT
POST	/api/settings/withholding	
GET	/api/events	60 days

LEDGER · UNDO 7

GET	/api/ledger	
POST	/api/ledger/edit	open BUY
POST	/api/ledger/delete	
GET	/api/realized	by year
GET	/api/chart-trades	
GET	/api/undo	peek
POST	/api/undo	apply

WATCHLIST · ALERTS 7

GET	/api/watchlist	
POST	/api/watchlist/add	
POST	/api/watchlist/remove	
GET	/api/alerts	met?
POST	/api/alerts/add	price·line·ind
POST	/api/alerts/update	re-arm
POST	/api/alerts/remove	

CHART WORKSPACE 6

GET	/api/charts/drawings	
POST	/api/charts/drawings	
GET	/api/charts/layouts	
POST	/api/charts/current	
POST	/api/charts/layouts/save	
POST	/api/charts/layouts/delete	

TRADE JOURNAL 7

GET	/api/journal	+ R stats
POST	/api/journal/create	
POST	/api/journal/update	
POST	/api/journal/delete	
GET	/api/journal/image	
POST	/api/journal/image	
GET	/api/journal/csv	export

Full request and response shapes: `DOCS.md › HTTP API`.

§07 · UPSTREAM & CACHE

Five free feeds, one shock absorber.

Each provider is called only from `market.js` or `macro.js`, and only through `cached()`. If a feed goes down, the terminal shows the last good value instead of failing.

Yahoo Finance

Quotes, history, FX, news, sectors, earnings

`/v7/finance/quote`
`/v8/finance/chart`
`/v1/finance/search`

FRED

Yields, CPI, Core PCE, Fed funds

`fredgraph.csv`

World Bank

Annual country macro

`api.worldbank.org/v2`

Forex Factory

This week's high-impact USD/SGD releases

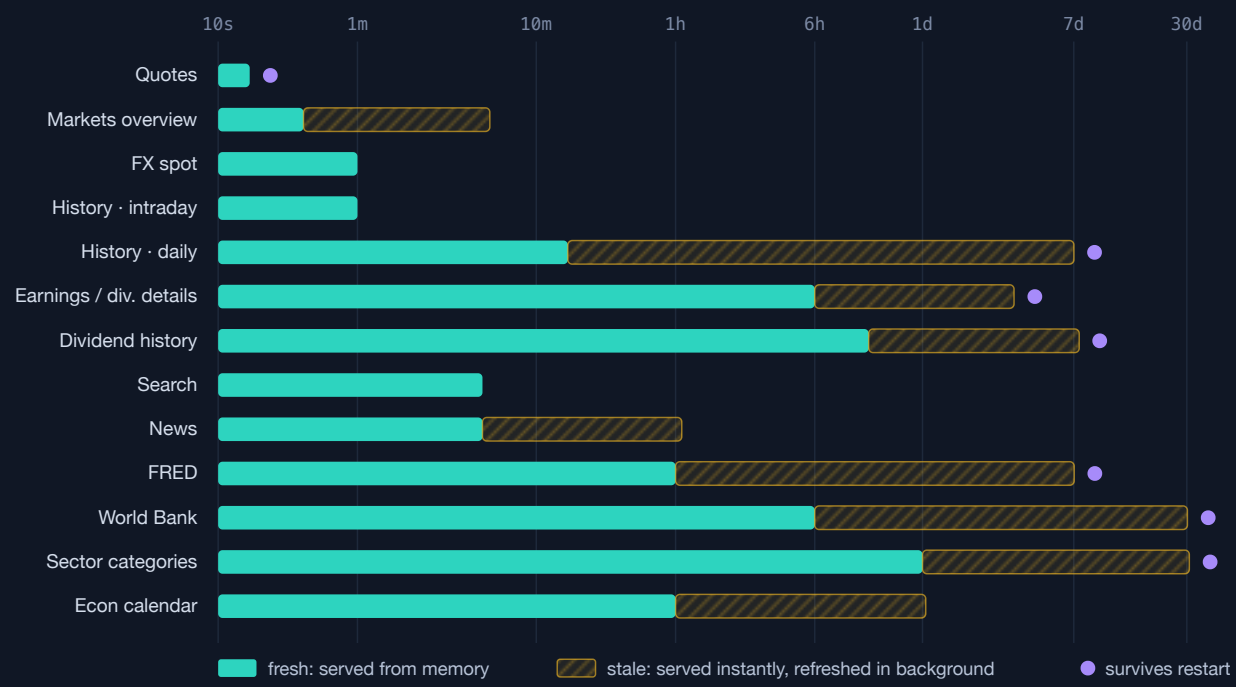
`ff_calendar_thisweek.json`

Federal Reserve

FOMC and dot-plot dates. Hard-coded, refreshed yearly

`fomccalendars.htm`

HOW LONG EACH ANSWER STAYS USEFUL



De-dupe

Concurrent callers share **one** in-flight request.

Backoff

HTTP 429 from both Yahoo hosts → pause **1 → 10 min**.

No range=max

Yahoo returns quarterly bars for it, so charts ask for **10y / 20y / 50y**.

Fair use

Personal use only. **Never redistribute** provider data. Yahoo's endpoint is unofficial.

§08 · THE MATHS

Did the stock make money, or the currency?

FinSight answers that for every position. Worked example (illustrative numbers): **10 shares bought at US$100 when USD/SGD was 1.30**. Now the price is US$120 and USD/SGD is 1.35.



UNREALIZED S$320 = STOCK 270 + FX 50

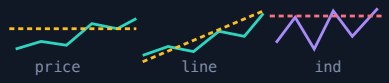
TWR Real returns

Replays every BUY, SELL and DELETE in date order. Values the book each trading day with **historical price and historical FX**. Daily returns are chained into TWR, and XIRR solves the dated cash flows. Compare against any benchmark, such as `^GSPC`.

VaR FX risk

Exposure by currency (foreign cash counts too). Shows the blended entry rate vs spot, a what-if shock slider and a **1-week VaR** at 1σ and 95% from weekly USD/SGD volatility. Per-currency risks are summed, which is conservative.

ALR Alerts



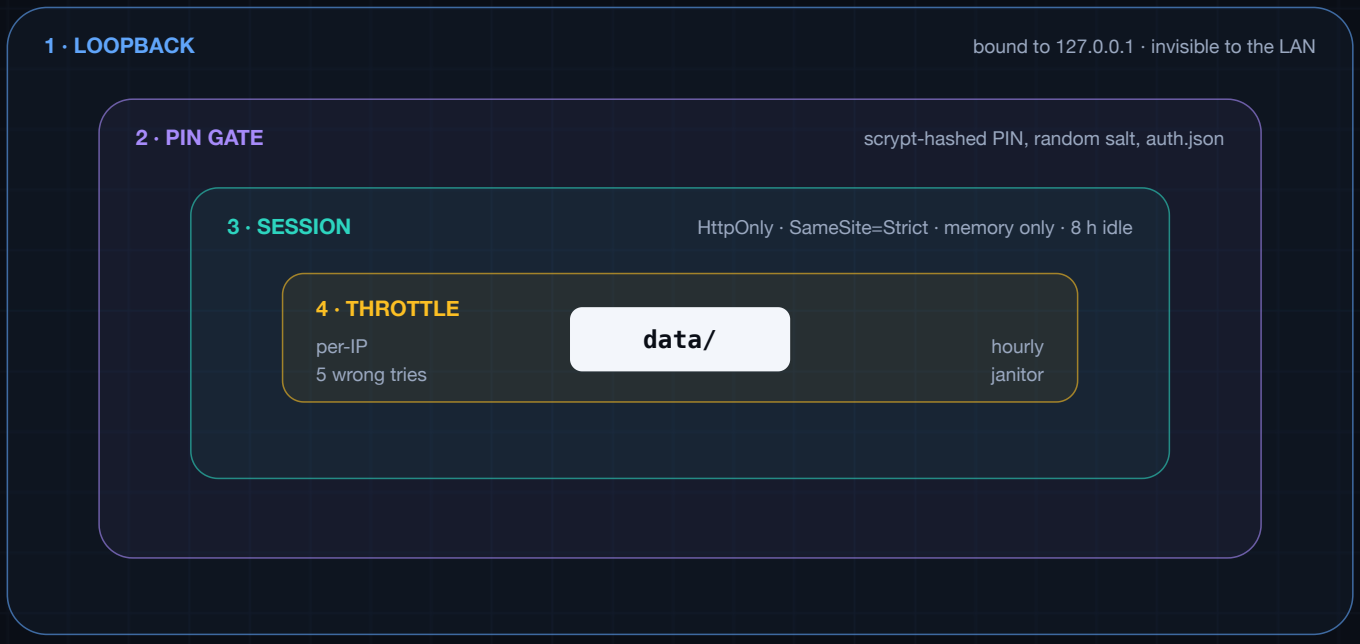
Checked server-side every 60 s with one desktop notification per trigger. Drawn lines are measured in **bar-index space**, so weekends don't bend them.

Dividends: net of the actual IBKR withholding when recorded, otherwise a per-market default (US 30%). Forward income uses Yahoo's forward rate or the trailing 12 months. Buys under about 7 days old show roughly zero FX P&L ("settling").

§09 · SECURITY & PRIVACY

Your data sits behind four walls.

To reach anything in `data/`, a request has to pass every ring. The UI shell is public. The data API is not.



WHAT FINSIGHT NEVER DOES

No secrets

No API keys, tokens or provider accounts exist to leak.

No telemetry

Upstream calls carry only symbols and series IDs. Never holdings.

No browser cache

The service worker stores static files only. Never `/api`.

No trading

No broker link. Alerts, strategies and the journal only record.

Static paths are normalised under `public/` to block traversal. A server restart clears all sessions. `data/` is git-ignored.

§10 · CONFIGURATION & OPERATIONS

From zero to terminal in one command.

10

No install step and nothing to compile. You need Node 18 or newer, then:

~/finsight — zsh

```
$ node server.js                      # or npm start · or double-click start.command
  running at  http://localhost:8000  (bound to 127.0.0.1)

$ PORT=9000 NOTIFY=0 node server.js    # override any env var inline

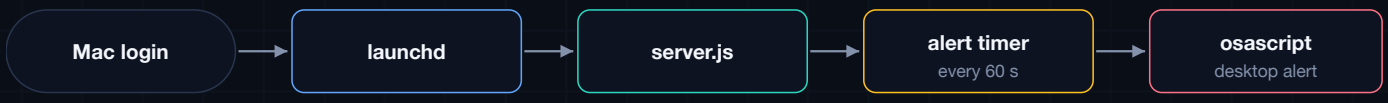
$ ./install-autostart.sh              # launchd agent com.finsight.terminal
  starts at login · restarts on crash · alerts fire with no browser open

$ npm test
  ℹ tests 96  ℹ pass 96  ℹ fail 0   # offline: fake Yahoo + temp data dir
```

DIALS · ENVIRONMENT VARIABLES

PORT **8000** HTTP port	HOST **127.0.0.1** bind interface	ALERT_INTERVAL **60 s** alert check, min 15	LOCK_IDLE_MIN **480** session idle, minutes
NOTIFY **on** 0 = no macOS popups	FINSIGHT_DATA_DIR **./data** where state lives	FINSIGHT_CACHE_PERSIST **on** 0 = memory-only cache	API KEYS **none** nothing to configure

BACKGROUND ALERT LOOP



First launch opens a wizard: set your name and base currency, then import a broker CSV, add holdings by hand or start empty. Back up by copying `data/`.

§11 · KNOWN GAPS

What it doesn't get right yet.

These come from the 30 Sep 2026 accounting review. Gaps 1, 4 and 5 were re-checked in today's source. In short: today it is a **securities + FX snapshot tracker**, not yet a reconciled whole-account ledger.

1 Cash and trades don't talk

A BUY doesn't reduce cash and a SELL doesn't credit it. Net worth can be overstated after a funded buy.

`lib/portfolio.js · buyUnlocked, setCash`

2 Re-import doubles up

Append mode adds the same statement again. Replace mode wipes history. Unsupported rows such as shorts drop silently.

`lib/portfolio.js · importCsv · lib/csv.js`

3 Two different "totals"

Performance replays securities only (no cash) and counts dividends gross. Benchmarks are price indices, not total return.

`lib/performance.js`

4 Missing FX becomes 1.0

A failed conversion uses 1:1 and is flagged in `fxMissing`, but the SGD total still looks like a real number.

`lib/portfolio.js · fx fallback`

5 Identity is just the ticker

The `ALL` view merges by symbol, so one ticker in two currencies merges. Quantity checks positivity, not finiteness.

`lib/store.js · viewPositions`

6 Soft defaults

XIRR doesn't flag multiple roots. WHT assumes a Singapore resident. Short FX history gives 0 volatility. Weights exclude cash.

`lib/returns.js · lib/market.js`

FIX ORDER

DATA-001
strict intake boundary

LEDGER-001
cash-linked, idempotent import

RISK · BENCH · THESIS
on validated records

§12 · COMMANDS & EXTENSION HOOKS

Type it. Or extend it.

12

Everything in the UI can also be typed into the command bar. Every subsystem has one obvious place to plug in something new.

FINSIGHT › command bar

› ADD AAPL 10 195.50 2024-01-15	buy or average in · date captures that day's FX
› SELL AAPL 5 [price]	sell · records realized stock + FX P&L
› DEL AAPL	remove the position (no sale recorded)
› UNDO	roll back the last change (25 deep)
› LEDGER	all trades · edit a buy · realized by year
› CASH USD 5000	set uninvested cash (0 clears)
› WHT US 15	withholding rate for a market
› Q TSLA · TSLA	detail, chart and news
› WATCH NVDA · UNWATCH NVDA	watchlist
› ALERT AAPL > 320	price alert (< for below)
› NEWS TSLA · MKT · HELP	news · refresh markets · help

EXTENSION SOCKETS

DATA

New source → helper in `lib/` wrapped in `cached()`, a route in `server.js`, render in `app.js`, test with the fake market.

MUTATION

Wrap it as `withLock(() => withUndo('Label', …))`. It is then serialized and undoable.

INDICATOR

Pure function in `indicators.js`, registered in `IND` in `chart.js`. Use `pane:true` for a sub-pane.

DRAWING

Add to `TOOLS` and `GROUPS`, then handle it in `_drawOne`, `_hit` and `_handles`.

UI

Card → `CARD_DEFS` · panel → `PANEL_PIDS` · FRED/WB series → `<option>` in `index.html`.

MARKETS

Overview symbols → `MARKET_GROUPS` in `lib/market.js`.

TEST SUITES · NODE:TEST · 96 / 96 GREEN

cache · csv · ledger · portfolio · performance · returns · indicators · chartfeatures · strategies · strategy · journal